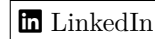
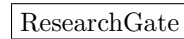


Onyero Walter Ofuzim

Calgary, AB, Canada (Open to relocate)
532 Cedar Crescent SW, Calgary, AB T3C 2Y8

[Email](#)[GitHub](#)[Portfolio](#)[LinkedIn](#)[ResearchGate](#)

+1 403 605 0955

Available immediately; M.Sc. defence expected March 2026.

Summary

Engineer and software-focused M.Sc. candidate in Electrical & Software Engineering with experience across battery modelling & control, data/analytics automation, and backend/API development. Hands-on with Python, SQL, Linux, testing/CI, and building reproducible pipelines. Proven in research environments and production-like projects: system identification, simulation, data quality, dashboards, and service-oriented designs. Open to relocate across Canada; available immediately.

Skills

Programming: Python, SQL, C/C++, MATLAB, Bash

Backend/Web: FastAPI/Flask (Python), REST APIs, WebSockets (project), Node.js/Express (project), React/TypeScript (project), Swagger/OpenAPI (project)

Data: pandas/NumPy/SciPy, PySpark/Spark SQL (projects), Delta Lake (projects), dbt/Airflow (project exposure), Power BI, Excel (advanced)

ML/AI: scikit-learn, TensorFlow, PyTorch, XGBoost (projects/coursework); evaluation, feature engineering, time-series basics

Cloud/DevOps: Git/GitHub, GitHub Actions, Docker, Kubernetes (projects), Linux tooling, CI/testing habits

Controls/Modeling: classical/state-space control, observers/Kalman (research), system identification (ARX/BJ/PEM), simulation/validation

Networking/Systems: TCP/IP, sockets/poll (project), SNMP concepts, Wireshark/tcpdump awareness

Testing: pytest, unittest, xUnit (project), API/UI automation (projects), basic quality practices

Education

M.Sc., Electrical & Software Engineering

Jan 2024 – Expected Mar 2026

University of Calgary

GPA: 3.85

Focus: Advanced Control & System Identification, AI/ML, Battery Modelling, Software Engineering Practices

B.Eng., Electrical/Electronic Engineering

2021

University of Benin, Nigeria

Focus: Embedded Systems, Power/Electronics Foundations, Control Fundamentals

Experience

Graduate Research Assistant / Teaching Assistant

Calgary, AB

Jan 2024 – Present

University of Calgary — Alberta Innovates (DICE) / TC Energy

- Built physics-based Li-ion models (SPMe/DFN) in Python (PyBaMM) and MATLAB; designed SOC/SOE observers (EKF/UKF) and validated against cycling data.
- Developed parameter estimation pipelines (nonlinear least squares, PSO) for kinetics/diffusion/transport; ran sensitivity and Monte Carlo studies to quantify uncertainty.
- Implemented degradation mechanisms (SEI growth, LAM, resistance rise) and coupled them to state equations for life-cycle prediction and health trend analysis.
- Built hybrid data+physics estimators for degradation trajectories using ARX/BJ/NARX and LSTM baselines; generated early-warning indicators and basic RUL signals from health features.

- Authored requirements, experiment logs, and verification plans; created scenario suites, assertions, and unit tests; maintained traceability with Git.
- Teaching Assistant for controls/optimization/software practices/computer organization; designed labs, mentored students on Git/testing/clean design.

Lead Technology Specialist (Analysis) / Network Support Analyst Lagos, Nigeria Mar 2023 – Nov 2023

MTN Nigeria — Network Performance Quality Assurance (NPQA)

- Used Python and analytics tooling to extract insights from network KPI datasets; delivered actionable recommendations for performance optimization and incident reduction.
- Built scripts/automations to streamline recurring operational tasks; reduced manual effort and improved turnaround time for reports.
- Worked extensively with terminals and Linux tooling; supported SNMP-related workflows and fault/performance troubleshooting.
- Collaborated cross-functionally to integrate software-driven solutions into operational routines; improved visibility into recurring issues.

Network Engineer Intern

Lagos, Nigeria

Aug 2022 – Feb 2023

MTN Nigeria — NPQA (Core Performance)

- Supported network performance workflows; analyzed operational data using Excel/Power BI and assisted in root cause reporting.
- Assisted troubleshooting and monitoring activities; contributed to documentation and knowledge transfer for recurring issues.

Graduate Research Assistant (Embedded/IoT)

Benin City, Nigeria

Jan 2020 – Aug 2021

University of Benin, Dept. of Electrical/Electronic Engineering

- Prototyped GSM-enabled battery monitoring/control device; contributed to embedded systems + IoT research for energy sustainability.
- Co-authored publication on IoT-based battery monitoring/control. [ResearchGate](#)

Network Operations/Support Engineer Intern

Lagos, Nigeria

Nov 2019 – May 2020

MTN Nigeria — NSMC

- Rotated across TX/IP MPLS, Data & Internet Services, and NSS units; monitored capacity/performance and supported escalation workflows.
- Contributed to SNMP monitoring support and operational reporting; used Excel for daily/weekly/monthly reporting and RCA tracking.

Automation/Electrical Engineer Intern

Lagos, Nigeria

Aug 2019 – Nov 2019

Nigerian Bottling Company (Coca-Cola HBC)

- Supported troubleshooting and inspection of electrical/automation systems; improved maintenance documentation and reporting flow.

Open-Source Contributions

Meshery (Layer5) — Contributor

Dec 2025 – Present

Linux Foundation LFX Mentorship (Remote)

- Participated in community working groups and onboarding; scoped starter issues across Meshery Server/Operator (Go) and UI (React/Next/GraphQL).
- Set up local dev workflow, linting/testing; focused on small fixes, documentation, and issue triage aligned with contributing guidelines.

Completed Projects and Job Simulations (Selected)

- **Kinematic Analysis of Mecanum Robot (Co-author)** — Derived wheel-to-chassis mapping and control constraints; built CoppeliaSim model. [ResearchGate](#)
- **PEM Electrolyzer Battery Data Prediction Using NARX (Paper)** — Built NARX forecasting with FNN and LSTM-based architectures for voltage/current time-series; evaluated predictive performance for maintenance insights. [ResearchGate](#)
- **Olist E-Commerce Medallion Pipeline (PySpark + Delta)** — Built Bronze/Silver/Gold lakehouse; engineered Gold marts (SLA KPIs, revenue by region, RFM features) and exported Parquet for BI. [GitHub](#)
- **Mobility Lakehouse (Polyglot, Dockerized)** — Local end-to-end lakehouse: Kafka → Spark/Delta Bronze → curated marts; orchestration + data quality; metrics via Prometheus/Grafana; UI catalog. [GitHub](#)
- **OSS Lakehouse in Docker (Spark + Delta + MinIO + MLflow)** — Reproducible local stack enabling Bronze/Silver/Gold and MLflow tracking; designed for repeatable demos and validation tooling. [GitHub](#)
- **Synapse-Style SQL on Files (DuckDB + Spark SQL)** — Practiced querying files-in-place (CSV/JSON/Parquet) partition pruning, CETAS-style transforms, and shared lake folder layouts. [GitHub](#)
- **Banking System (ASP.NET Core Web API + EF Core + xUnit)** — Modular, production-style backend with clean architecture patterns, persistence, Swagger/OpenAPI, and tests. [GitHub](#)
- **Order Notifications Demo (FastAPI + WebSockets + React/TS)** — Real-time order status updates via WebSocket channel; REST API for create/update; frontend receives live events. [GitHub](#)
- **Clinic Inbox Agent (FastAPI + SQLite + Next.js)** — Simulated AI-assisted triage dashboard: categorization/urgency and suggested actions; staff can confirm/override; designed as LLM-ready. [GitHub](#)
- **API Smoke Tests (pytest + requests)** — **reqres.in** — CI-friendly API smoke pack with retries, negative tests, and lightweight schema checks; produces reports for quick health signals. [GitHub](#)
- **UI E2E Automation — SauceDemo (Behave + Selenium)** — End-to-end purchase flow using Page Object Model; runs in GitHub Actions; structured for maintainable UI automation. [GitHub](#)
- **Cisco-Style Multi-Client TCP Echo Server (C++ / BSD Sockets)** — Built multi-client server with `poll()`, partial read/write handling, and line-based protocol behaviors. [GitHub](#)
- **SNMP Network Health Dashboard (FastAPI + SQLite)** — Periodic polling + storage + JSON APIs + simple dashboard; clean separation between collector, client, API, and DB models. [GitHub](#)
- **kubeops-fastapi-operator (KOPF)** — Kubernetes operator managing FastAPI + Postgres deployments using CRDs and reconciliation loops; includes tests/runbooks. [GitHub](#)
- **Kubernetes Cost Visibility Lab (OpenCost + Grafana + Kyverno)** — Enforced cost labels, deployed monitoring stack, and exposed allocation data through a FastAPI proxy. [GitHub](#)
- **Cafeteria Menu RAG Assistant (FastAPI + pgvector + Streamlit)** — Document ingestion/chunking, embedding, vector search, and grounded Q&A; containerized with configurable embedding modes. [GitHub](#)
- **Supplier Quality & Capability Dashboard (Power BI)** — Single-page operational view: defect rate ranking, Cp/Cpk trends, NCR backlog health, and actionable thresholds for weekly triage. [GitHub](#)

Job Simulations (Selected)

- **J.P. Morgan Software Engineering (Forage)** — Environment setup, data visualization tasks using Perspective; debugging and clean output validation.
- **Citi ICG Technology (Forage)** — UML/state modelling, system enhancements, and research recommendations for ML credit risk assessment.
- **HPE Software Engineering (Forage)** — Proposed REST service design; built Spring Boot service and unit tests.
- **Lyft Back-End Engineering (Forage)** — Refactored and extended codebase using TDD; produced UML class diagram for improved architecture.
- **Ford EV Engineering (Forage)** — PID tuning analysis and cost-benefit evaluation for EV-related components.

Publications

- Proton Exchange Membrane Electrolyzer Battery Data Prediction Using NARX (Nonlinear AutoRegressive with Exogenous Inputs) Neural Network Model. [ResearchGate](#)
- Design and Construction of a Remote Battery Monitoring and Control Device Using IoT. [ResearchGate](#)

Licenses and Certifications

Data & Cloud

- DeepLearning.AI / AWS Data Engineering Professional Certificate — DeepLearning.AI Sep 2025
- Snowflake Data Engineering Professional Certificate — Snowflake Sep 2025
- Docker & Kubernetes Masterclass (Beginner → Advanced) — Packt Sep 2025
- Mastering Azure Databricks for Data Engineers Specialization — Packt Jul 2025
- Data Engineer in Python (Career Track) — DataCamp Nov 2024
- Data Analyst in SQL (Career Track) — DataCamp May 2023
- Introduction to Power BI — DataCamp Feb 2023
- Big Data Fundamentals with PySpark — DataCamp Sep 2022
- Introduction to PySpark — DataCamp Aug 2022
- Introduction to Airflow in Python — DataCamp Aug 2022
- Introduction to AWS Boto in Python — DataCamp Aug 2022
- AWS Cloud Concepts — DataCamp Aug 2022
- Database Design — DataCamp Aug 2022
- Intermediate SQL Server — DataCamp Aug 2022
- Intro to Relational Databases — DataCamp Aug 2022
- Intro to SQL Server — DataCamp Aug 2022
- PostgreSQL Summary Stats & Window Functions — DataCamp Jun 2022
- Intermediate SQL — DataCamp Jun 2022
- SQL Fundamentals (Skill Track) — DataCamp Jun 2022
- Joining Data in SQL — DataCamp May 2022

Machine Learning & AI

- Machine Learning — Stanford Online Dec 2021
- Deep Learning with PyTorch: GAN — Coursera Sep 2022
- Fine-Tune BERT for Text Classification (TensorFlow) — Coursera Sep 2022
- Aerial Image Segmentation with PyTorch — Coursera Sep 2022
- Object Localization with TensorFlow — Coursera Sep 2022
- Linear Regression with NumPy and Python — Coursera Sep 2022
- Tesla Stock Price Prediction (Prophet) — Coursera Sep 2022
- Tweet Emotion Recognition with TensorFlow — Coursera Sep 2022
- Build your first AI Stock Predictor (Amazon Forecast) — Coursera Jan 2022
- Hyperparameter Tuning with Keras Tuner — Coursera Jan 2022
- Build your first ML Pipeline (Dataiku) — Coursera Dec 2021
- Supervised Learning with scikit-learn — DataCamp Jul 2022
- Extreme Gradient Boosting with XGBoost — DataCamp Jul 2022
- Statistical Thinking in Python (Part 1 & 2) — DataCamp May 2022
- Unsupervised Learning in Python — DataCamp May 2022
- Dimensionality Reduction in Python — DataCamp Jul 2022

Software, Programming & Engineering Practices

- Open Source Software Development, Linux and Git Specialization — Linux Foundation Jul 2023

- Programming with JavaScript — Meta Oct 2023
- Programming in Python — Meta Aug 2023
- Databases for Back-End Development — Meta Sep 2023
- Database Structures & Management with MySQL — Meta Aug 2023
- Advanced MySQL Topics — Meta Aug 2023
- Version Control — Meta Jun 2023
- Programming Fundamentals in Kotlin — Meta Jun 2023
- Create an Android App with Kotlin — Coursera Jul 2023
- Developing Python Packages — DataCamp Jun 2022
- Object-Oriented Programming in Python — DataCamp Jun 2022
- Python Programmer Track (Career Track) — DataCamp Jun 2022

Robotics & Autonomous Systems

- Robotics (Aerial, Motion Planning, Legged Mobility, Perception) — University of Pennsylvania Oct 2022 – Mar 2023
- Cooperation in Autonomous Vehicles — IEEE Oct 2022
- Developing & Validating Control Systems for Connected & Automated Vehicles — IEEE Oct 2022
- Self-Driving Car Fundamentals: Featuring Apollo — Udacity Feb 2022
- Introduction to Self-Driving Cars — University of Toronto Jan 2022
- Modern Robotics, Course 1: Foundations of Robot Motion — Northwestern University Dec 2021

Analytics / BI & Excel

- Excel Basics for Data Analysis — IBM Jul 2022
- Excel Essentials for Data Analytics (IBM Badge) — IBM Jul 2022

IBM / Data Foundations

- Python for Data Science and AI (IBM Badge) — IBM Aug 2022
- Python for Data Science, AI & Development — IBM Aug 2022
- Introduction to Data Analytics — IBM Apr 2022
- Data Analytics Essentials (IBM Badge) — IBM Apr 2022

Additional Information

Relocation: Open to relocate across Canada

Availability: Immediately available; M.Sc. defence expected March 2026